

## PROJECT NAME: SECURITY MONITORING WITH SPLUNK.

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STATUS: in progress-80% completed

EXPECTED COMPLETION: before the end of the month.

WHAT IS REMAINING: just to add images to the already answered question.

## THE SOC PROJECT QUESTIONS IS BELOW

These questions focus on the core concepts of Splunk, including basic search, filtering, and understanding key fields. (TO BE SUBMITTED WITH SCREENSHOTS )

Scenario: Website Traffic Analysis 🌐

Your web server logs are indexed in Splunk. You want to see all traffic from a specific IP address 192.168.1.100.

Question: Write a search query to find all events with the clientip field equal to 192.168.1.100.

Scenario: Error Monitoring ⚠️

You're a system administrator and want to quickly find all critical errors in the last 24 hours. The logs use the terms "ERROR" and "FAILURE" to indicate critical issues.

Question: Construct an SPL query to search for events containing either "ERROR" or "FAILURE" within the \_internal index.

Scenario: User Activity Tracking 👤

Your company's authentication logs are being monitored. You need to investigate a user named "jdoe" who may have attempted to log in multiple times unsuccessfully.

Question: Write a query that finds all events where the user field is "jdoe" and the status field is "failed".

Scenario: Identifying Top Spenders 💰

You have sales data for an e-commerce website. You want to identify the top 10 customers based on the total\_revenue field.

Question: Use the top command to list the top 10 values for the customer\_id field, sorted by the total\_revenue.

Intermediate SPL Exercises

These questions require students to use transforming commands to manipulate and present data in meaningful ways.

Scenario: Calculating Daily Average Response Time ⌚

You're tasked with monitoring the performance of a web application. You have a `web_traffic` index with a field `response_time_ms`.

**Question:** Write an SPL query to calculate the average `response_time_ms` for each day over the past week and display it as a timechart.

**Scenario: Counting Event Types by Host** 

You need to create a summary of the different types of events (`event_type`) occurring on each host.

**Question:** Use the `stats` command to count the number of events for each `event_type` and group the results by host.

**Bonus:** How would you display this data in a table with a column for host and a column for the count of each `event_type`?

**Scenario: Mapping IP Addresses to Geographic Locations** 

You want to visualize the geographic origin of web attacks. Your logs have a `src_ip` field.

**Question:** Use the `iplocation` command to get the geographic information (city, country, etc.) for each `src_ip` in your events. Then, display the top 5 countries from which the attacks are originating.

**Scenario: Finding Correlated Events** 

A user, "admin\_user," is reporting a series of login failures followed by a successful login.

You want to see the entire sequence of events for that user.

**Question:** Use the `transaction` command to group all login events for the user "admin\_user" and display them as a single transaction.

### Advanced SPL Exercises

These questions challenge students with more complex scenarios that combine multiple commands and advanced concepts.

**Scenario: Detecting a Brute-Force Attack** 

You suspect a brute-force attack is targeting your login page. You have a log of failed login attempts (`action="login_failure"`). A brute-force attack is defined as more than 10 failed login attempts from the same IP address (`src_ip`) within a 1-minute window.

**Question:** Write an SPL query that identifies all IP addresses that meet the brute-force criteria.

**Scenario: Creating a Dashboard Panel** 

Your manager needs a dashboard panel that shows the number of unique users (`unique_users`) who accessed a specific application (`app="sales_portal"`) each day for the last 30 days.

**Question:** Create a search query that calculates the count of unique users per day and is suitable for a timechart visualization.

**Scenario: Data Enrichment with a Lookup Table** 

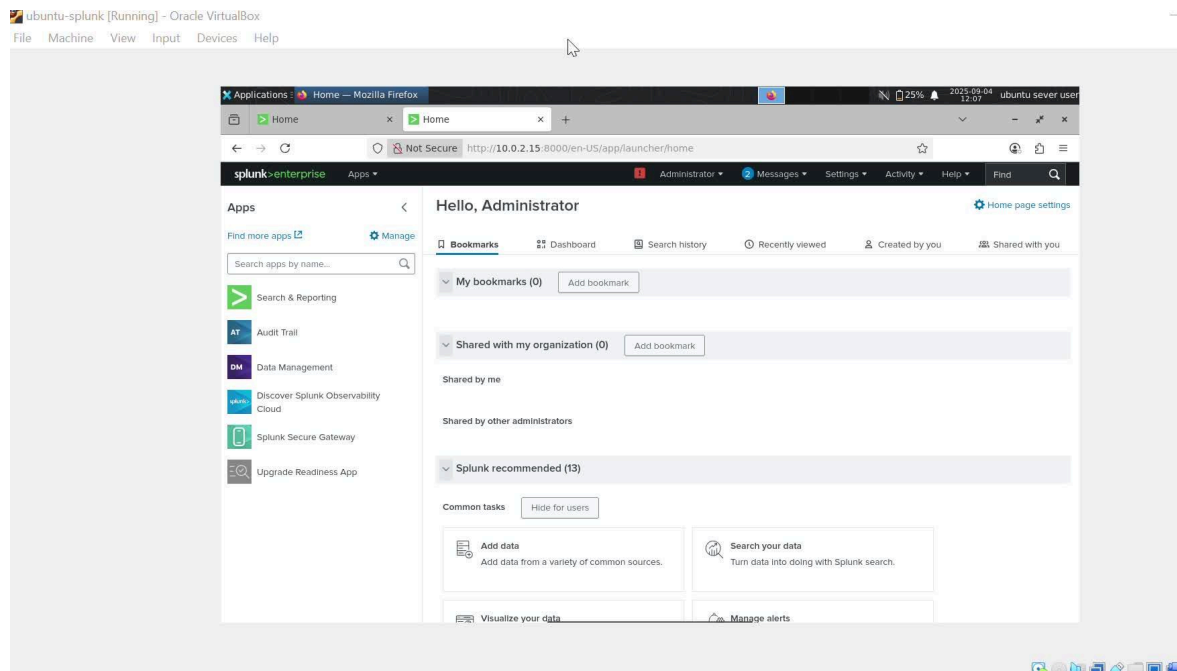
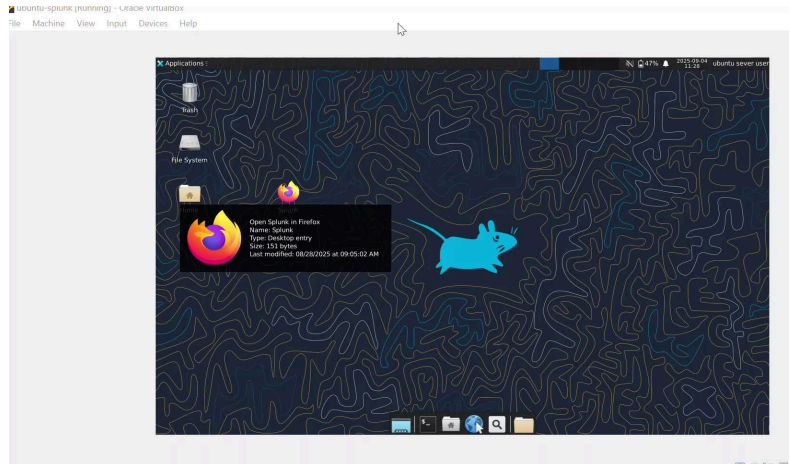
You have a CSV file (`threat_list.csv`) with known malicious IP addresses. You want to use this list to identify any events in your web logs that originate from a malicious IP.

**Question:** Write a query that uses a `lookup` command to compare the `src_ip` field in your web logs against the `threat_list.csv` file and only returns events where a match is found.

IN THIS PROJECT I WILL BE MAKING USE OF THREE VMS

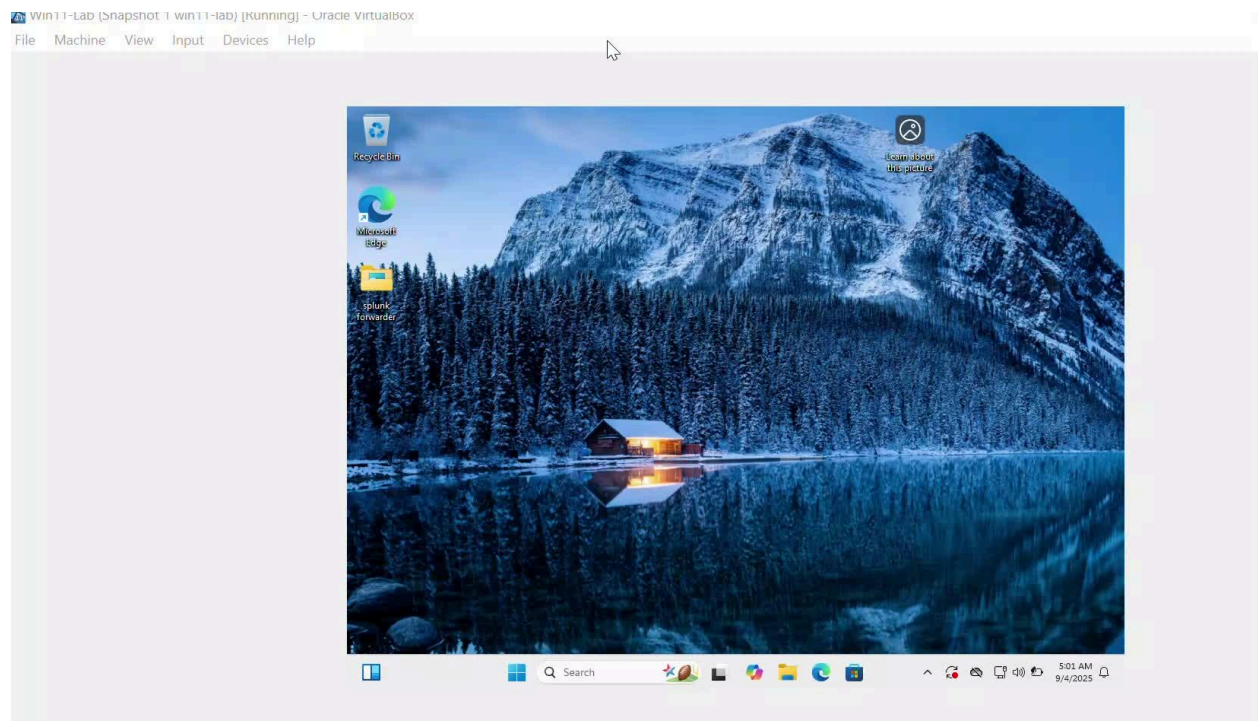
## 1. UBUNTU-SERVER

REASONS: BEING THAT IT IS LIGHT WIGHT, STABLE, AND WIDELY SUPPORTED, making it perfect for running splunk without wasting resources. UBUNTU-SERVER(SPLUNK)



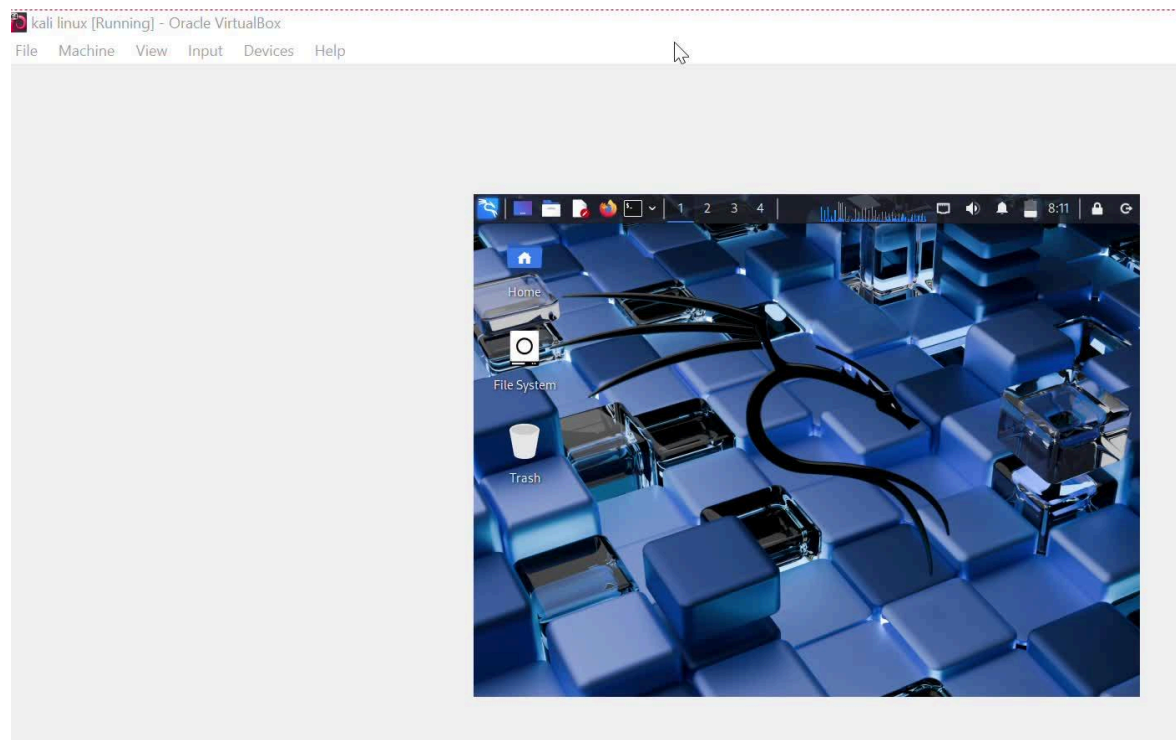
## 2. Win11-lab

REASONS: BEING THAT IT HOST MY SPLUNK FORWARDER, LETTING ME SAFELY COLLECT AND FORWARD LOGS FOR ANALYSIS TO MY SPLUNK



### 3. KALI LINUX

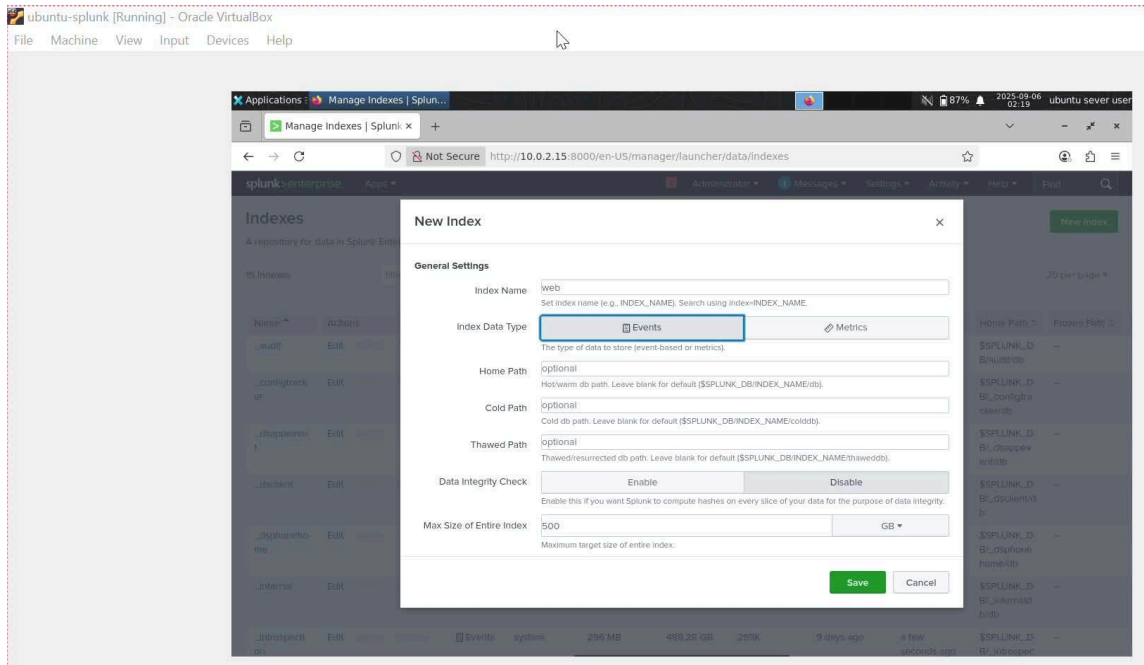
REASONS: BEING THAT I CAN BE ABLE TO CARRY OUT BRUTE FORCE ATTACK



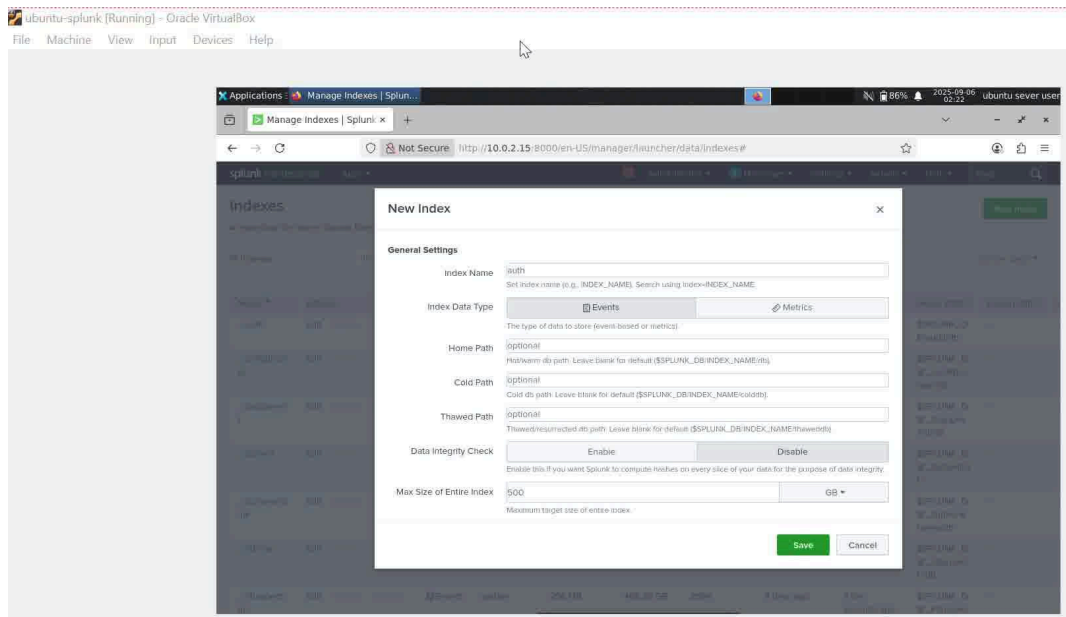
# CREATE INDEXES INSIDE SPLUNK.

GET READY WITH ME TO CREATE NEW INDEXES INSIDE MY SPLUNK : SUCH AS WEB (FOR WEB LOG), AUTH (FOR LINUX AUTH), WINEVENTLOG (FOR WIN EVENTS), SALES (FOR E-COMMERCE). WEB\_TRAFFIC (FOR PERFORMANCE), ATTACKS (WHICH IS OPTIONAL).

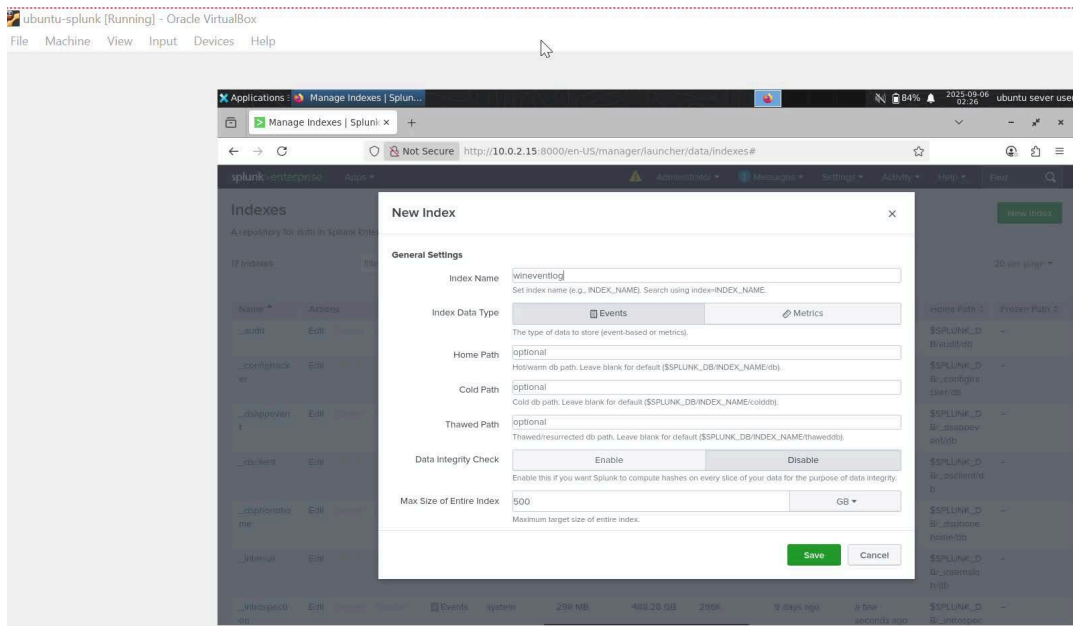
## 1. Web (for web log)



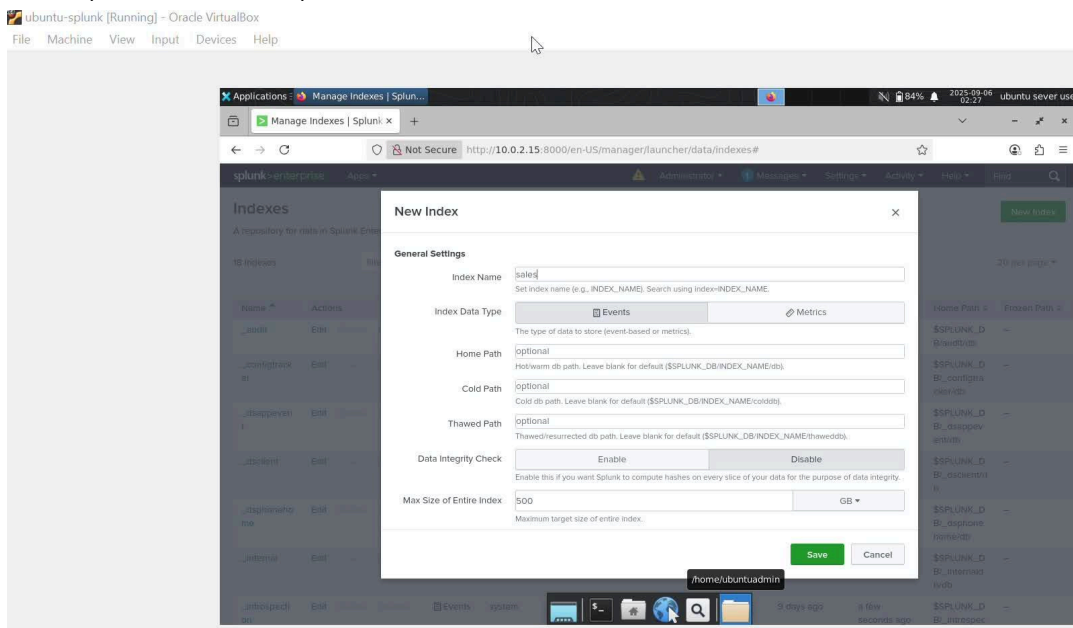
## 2. Auth (for linux auth)



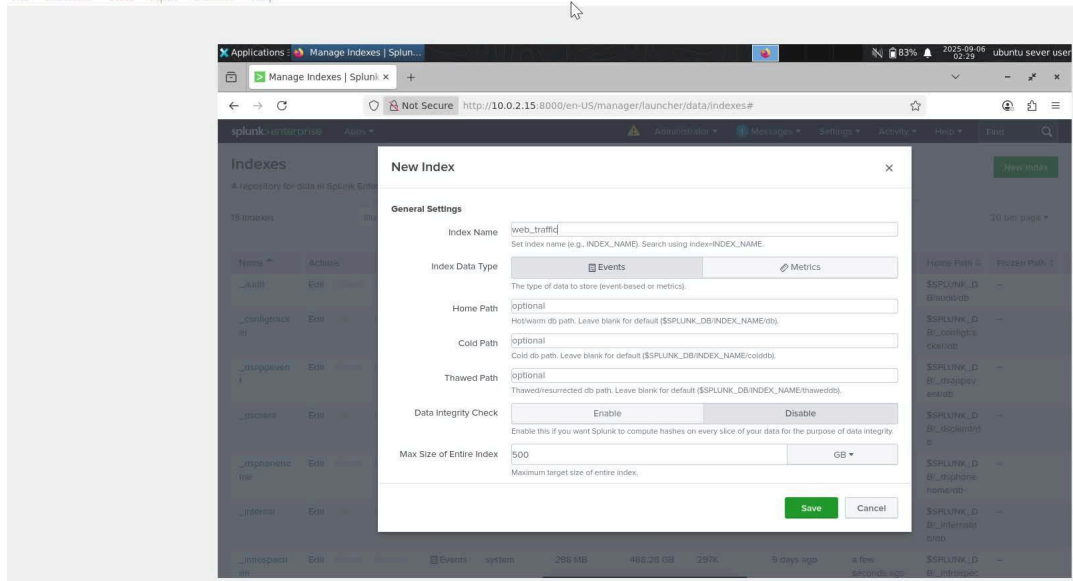
### 3. Wineventlog (for win events)



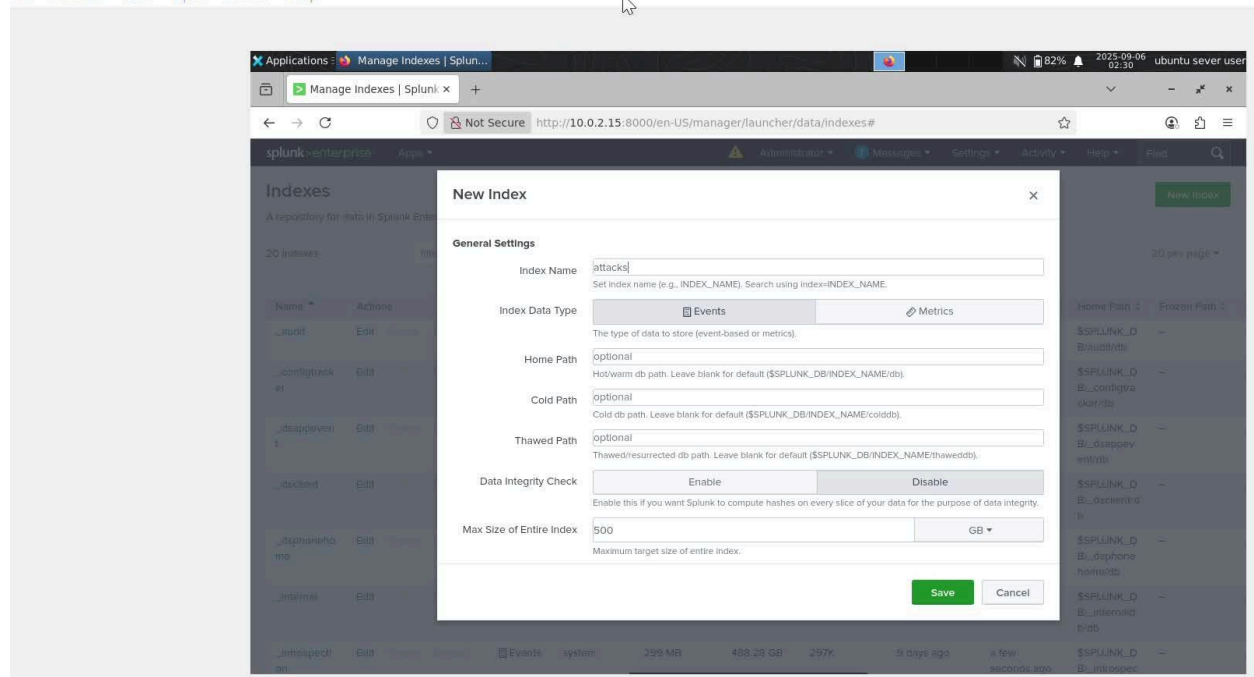
### 4. Sales (for e-commerce)



### 5. Web\_traffic (for performance)



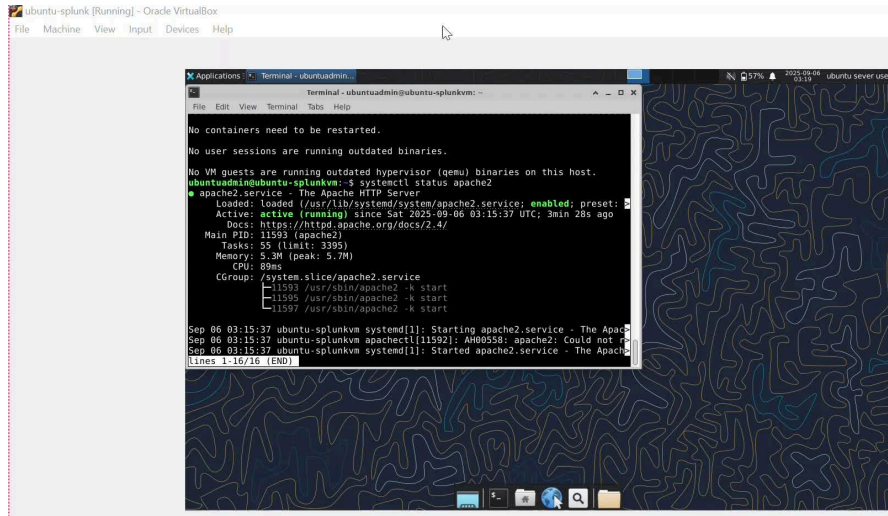
## 6. Attacks



## DATA ONBOARDING INTO SPLUNK.

- APACHE/NGINX access logs

First of all, i realised i do not have apache2. So have have to install it inside my ubuntu

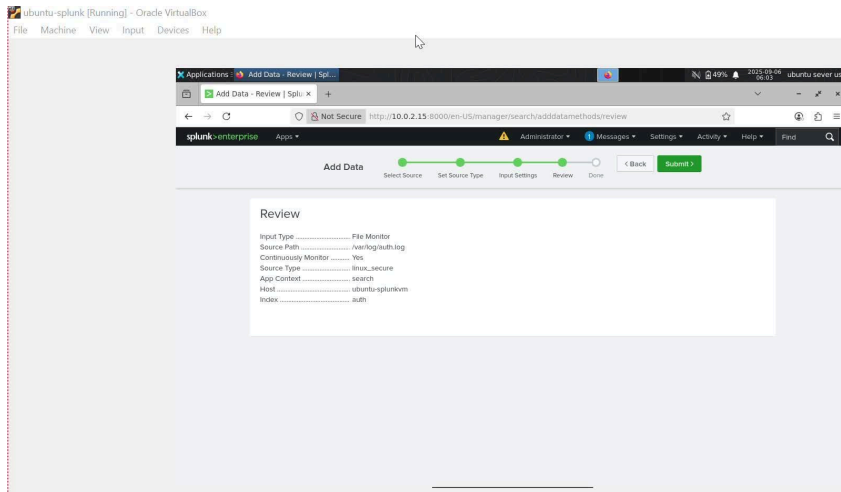


After that was created i went ahead to add data, monitor, files and directories to my /var/log/apache2/access.log

### REVIEW

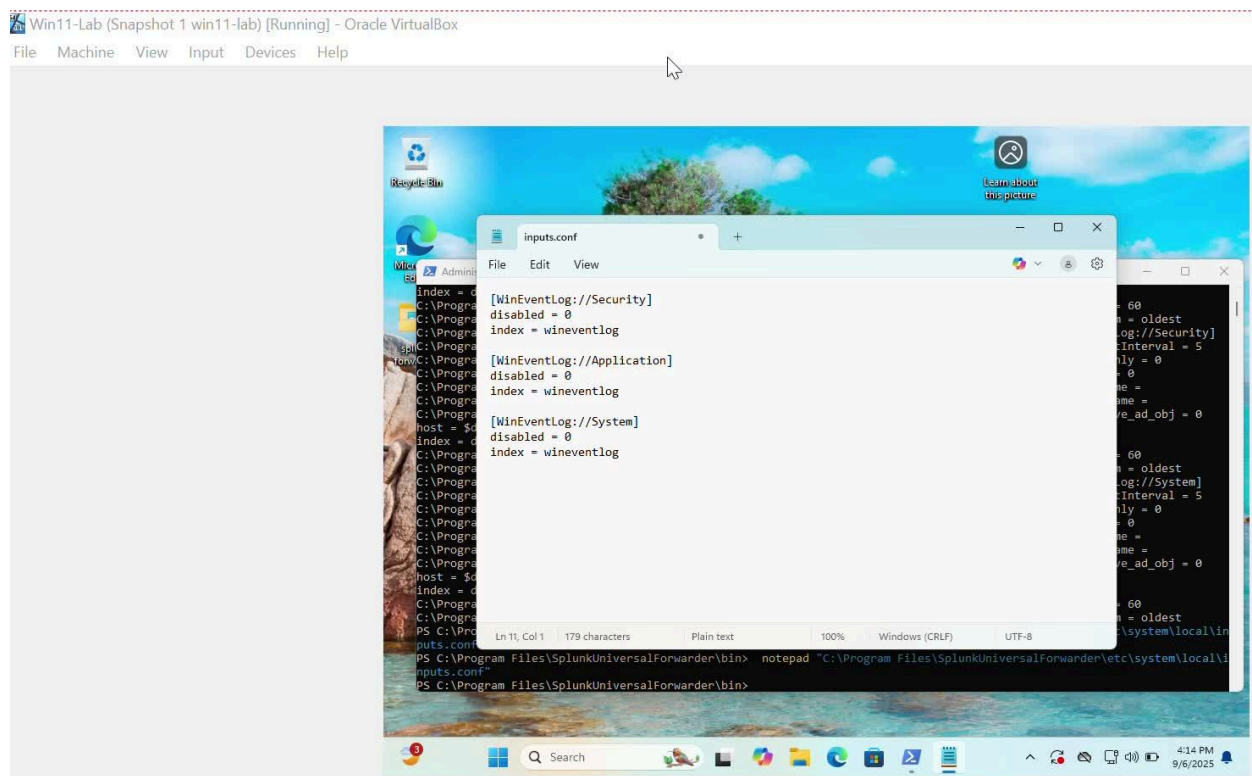
|                      |                             |
|----------------------|-----------------------------|
| Input Type           | file monitor                |
| Source Path          | /var/log/apache2/access.log |
| Continuously monitor | yes                         |
| Source Type          | access_combined             |
| App Content          | search                      |
| Host                 | ubuntu-splunkvm             |
| Index                | web                         |

- LINUX AUTH LOGS



- WINDOW EVENT LOG

Making use of splunk universal forwarder, inside my win11-lab to send win-event-logs such as Security, Application, System to splunk.



Now let me show you how my WinEventLog look like inside my splunk.  
Mind you my win11-lab, hostname is DESKTOP-IH53T40 and my index is set to default or main

So Typing: `index=* host="DESKTOP-IH53T40" sourcetype="WinEventLog:Application"` in the search and reporting will give you they pictures below

ubuntu-splunk [Running] - Oracle VirtualBox  
File Machine View Input Devices Help

The screenshot shows the Splunk Search & Reporting interface. The search bar contains the query: `index=* host="DESKTOP-IH53T40" sourcetype="WinEventLog:Application"`. The results show 224 events. The timeline visualization shows a series of green bars representing event frequency over time. The table view shows the following data:

| Time                  | Event                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9/6/25 2:05:51.000 PM | 09/06/2025 07:05:51 AM<br>LogName=Application<br>EventCode=256<br>EventType=4<br>ComputerName=DESKTOP-IH53T40<br>Show all 15 lines<br>host = DESKTOP-IH53T40 source = WinEventLog:Application sourcetype = WinEventLog:Application |
| 9/6/25 1:40:42.000 PM | 09/06/2025 06:40:42 AM<br>LogName=Application<br>EventCode=16384<br>EventType=4<br>ComputerName=DESKTOP-IH53T40                                                                                                                    |

So Typing: `index=* host="DESKTOP-IH53T40" sourcetype="WinEventLog:Security"` in the search and reporting will give you they picture below

ubuntu-splunk [Running] - Oracle VirtualBox  
File Machine View Input Devices Help

The screenshot shows the Splunk Search & Reporting interface. The search bar contains the query: `index=* host="DESKTOP-IH53T40" sourcetype="WinEventLog:Security"`. The results show 1,135 events. The timeline visualization shows a series of green bars representing event frequency over time. The table view shows the following data:

| Time                  | Event                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9/6/25 2:05:26.000 PM | 09/06/2025 07:05:26 AM<br>LogName=Security<br>EventCode=4672<br>EventType=8<br>ComputerName=DESKTOP-IH53T40<br>Show all 31 lines<br>host = DESKTOP-IH53T40 source = WinEventLog:Security sourcetype = WinEventLog:Security |
| 9/6/25 2:05:26.000 PM | 09/06/2025 07:05:26 AM<br>LogName=Security<br>EventCode=4624<br>EventType=8<br>ComputerName=DESKTOP-IH53T40                                                                                                                |

The screenshot displays the Splunk 10.0.0 web interface. At the top, there's a navigation bar with 'Applications', 'Search | Splunk 10.0.0', and a user profile 'ubuntu-sever user'. Below this is a search bar with the query 'index=\* host=DESKTOP-IH53T48\* sourcetype=WinEventLog:System\*'. The search results show two events from 9/6/25 at 07:05:25 AM. The interface includes a top navigation bar, a search bar, and a results table.

| Time               | Event                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9/6/25 07:05:25 AM | LogName=System<br>EventCode=17<br>EventType=4<br>ComputerName=DESKTOP-IH53T48<br>Show all 15 lines<br>host = <b>DESKTOP-IH53T48</b>   source = <b>WinEventLog:System</b>   sourcetype = <b>WinEventLog:System</b> |
| 9/6/25 07:05:25 AM | LogName=System<br>EventCode=17<br>EventType=4<br>ComputerName=DESKTOP-IH53T48<br>Show all 15 lines<br>host = <b>DESKTOP-IH53T48</b>   source = <b>WinEventLog:System</b>   sourcetype = <b>WinEventLog:System</b> |

- Using SSH brute attempts (reasons being that it is safe to use inside vms)

LET CHECK IN THE EVENT LOG IN SPLUNK

The screenshot shows the Splunk Enterprise web interface running in a browser. The search bar contains the query `index=auth invaliduser`. The search results show 14 events. The interface includes a sidebar with navigation options like Search, Analytics, Datasets, Reports, Alerts, and Dashboards. The main content area displays a list of events with columns for Time and Event. The events are filtered by the search query and show details about failed login attempts for the user 'invaliduser' from IP 192.168.56.104.

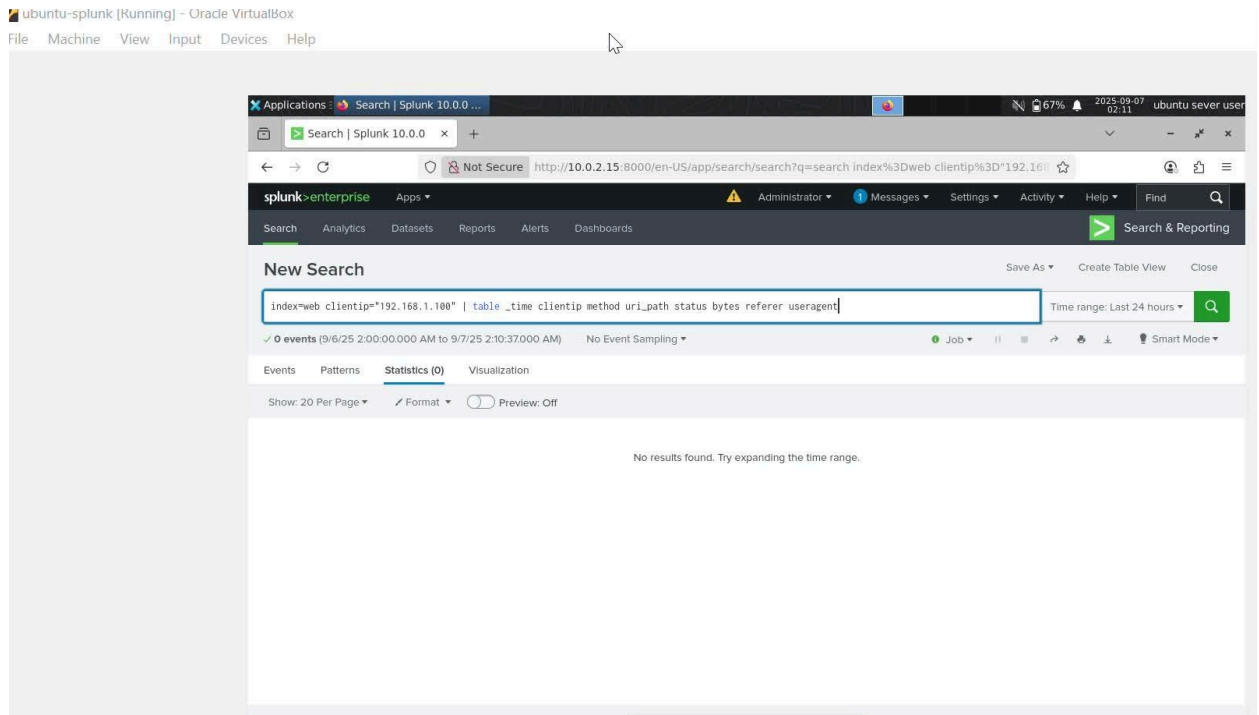
| Time                                                      | Event                                                                                                                                                                                       |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-09-07T01:38:03.371533+00:00<br>9/7/25 1:38:03.371 AM | ubuntu-splunkvm sshd[5992]: Invalid user invaliduser from 192.168.56.104 port 36592<br>host = ubuntu-splunkvm source = /var/log/auth.log sourcetype = linux_secure                          |
| 2025-09-07T01:36:47.036297+00:00<br>9/7/25 1:36:47.036 AM | ubuntu-splunkvm sshd[5820]: Failed password for invalid user invaliduser from 192.168.56.104 port 46100 ssh2<br>host = ubuntu-splunkvm source = /var/log/auth.log sourcetype = linux_secure |
| 2025-09-07T01:36:15.576780+00:00<br>9/7/25 1:36:15.576 AM | ubuntu-splunkvm sshd[5820]: Failed password for invalid user invaliduser from 192.168.56.104 port 46100 ssh2<br>host = ubuntu-splunkvm source = /var/log/auth.log sourcetype = linux_secure |
| 2025-09-07T01:35:21.843759+00:00<br>9/7/25                | ubuntu-splunkvm sshd[5820]: Invalid user invaliduser from 192.168.56.104 port 46100<br>host = ubuntu-splunkvm source = /var/log/auth.log sourcetype = linux_secure                          |

## PROJECT QUESTIONS COMPLETED.

1. Web traffic from a specific client ip 192.168.1.100. Due to the fact that i do not have any of my vms that use this specific ip or a data set, i could not generate a web traffic log for it.

Spl:

`index=web clientip="192.168.1.100" | table _time clientip method uri_path status`



This is the web traffic from client ip 192.168.56.102. The ip belong to my win11-lab(vm). I was able to generate web traffic log for

## 2. ERROR MONITORING- CRITICAL ERRORS LAST 24h in \_internal

Spl

index=\_internal earliest=-24h "ERROR" OR "FAILURE"

THE SPL is not for forwarded logs but for splunk's self-monitoring logs

Sources include: splunkd.log, metrics.log, scheduler.log

Mind you in this spl you can see user login failed.

## 3. TO GENERATE THIS

PARTICULAR auth.log for user jdoe we can make use of kali

Inside The root terminal of kali, enter this command

Bash

For i in {1..20}; do ssh jdoe@<192.168.56.103> -o StrictHostKeyChecking=no done

Note: The ip you are seeing is for (ubuntu host only ip)

After inputting that command press enter

And kali will reply you with

Password for jdoe= you can put any keyword and it will generate failed password.

And each failed attempt is logged by ubuntu in /var/log/auth.log

THEN IN SPLUNK YOU MAKE USE OF SPL IN SEARCH AND REPORTING  
SPL

```
index=auth user=jdoe status=failed | table _time host user status src_ip
```

#### 4. TO IDENTIFY TOP SPENDERS - TOP 10 BY TOTAL\_REVENUE

To create a sales log we can use win11-lab to simulate sales events as text/csv logs then let us forward them to my ubuntu (splunk)

In win11-lab open powershell as an administrator, run this script

Powershell

```
# Fake sales log generator
```

```
$logPath = "C:\logs\sales.log"
```

```
New-Item -ItemType Directory -Force -Path (Split-Path $logPath)
```

```
for ($i=0; $i -lt 20; $i++) {
```

```
    $time = Get-Date -Format "yyyy-MM-dd HH:mm:ss"
```

```
    $cust = "C" + (Get-Random -Minimum 100 -Maximum 110)
```

```
    $item = Get-Random -InputObject "Laptop","Phone","Tablet","Headphones","Monitor"
```

```
    $revenue = Get-Random -Minimum 50 -Maximum 1500
```

```
    "$time customer_id=$cust item=$item total_revenue=$revenue" | Out-File -Append $logPath
```

```
}
```

This will create something like

```
2025-09-07 15:01:23 customer_id=C105 item=Phone total_revenue=350
```

```
2025-09-07 15:01:24 customer_id=C109 item=Laptop total_revenue=1200
```

```
2025-09-07 15:01:25 customer_id=C101 item=Tablet total_revenue=220
```

We have already configure splunk UF on Win11-lab before

So what next is to add an input in inputs.conf to monitor my sales log:

Inputs.conf

```
[monitor://C:\logs\sales.log]
```

```
index = sales
```

```
sourcetype = sales_log
```

Finally i will restart the UF service so it starts shipping to my ubuntu splunk.

Inside my splunk, i will click on search and reporting and i will type spl

Spl

```
index=sales | stats sum(total_revenue) as revenue by customer_id | sort - revenue | head 10
```

## **5. Daily average response time over past week timechar**

**index=web\_traffic** would come from different web servers or load balancers (Apache, Nginx, IIS, F5, HAProxy, etc.).

**Logs typically include:**

**Timestamp**

**Client IP**

**HTTP method + URL**

**Status code (200, 404, 500, etc.)**

**Response time in ms**

**Example (Nginx access log, custom format with response time):**

```
192.168.1.55 - - [07/Sep/2025:15:02:01 +0000] "GET /api/login HTTP/1.1" 200 512 "-"  
"Mozilla/5.0" 35
```

**The response time in ms. Should be the last number 35**

**Mind you Splunk UF on the web server forwards these logs into index=web\_traffic.**

**So, typing this search query inside splunk**

**Spl**

**Index=web\_traffic earliest= -7d@d | timechart span=1d avg(response\_time\_ms) as avg\_ms**

## **6. Count event type by host (+ bonus pivot table)**

**host = automatically set by the UF (Win11, Ubuntu, maybe Kali).**

**event\_type = either defined in your logs, created with Splunk's field extractions, or faked with eval for practice.**

**Spl**

**index=\* event\_type=\* | stats count by host event\_type**

**Tabular matrix (host as rows, event\_type as columns)**

**index=\* event\_type=\* | chart count over host by event\_type**

**I am suppose to Screenshot: Statistics (first SPL) and Visualization/Table (second SPL**

## **7. MAP IPS to geolocation and show top 10 countries**

**First of all in ubuntu let simulate logs that contain src\_ip based on the fact that their is no external traffic.**

**Let Make a log file with random IPs: in my ubuntu vm**

**cat <<EOF > /var/log/attack.log**

**2025-09-07 15:01:23 src\_ip=185.60.216.35 action=login\_failed**

**2025-09-07 15:01:24 src\_ip=203.0.113.42 action=scan**

**2025-09-07 15:01:25 src\_ip=8.8.8.8 action=login\_success**

**2025-09-07 15:01:26 src\_ip=91.198.174.192 action=exploit\_attempt**

**2025-09-07 15:01:27 src\_ip=202.54.1.5 action=ddos**

**EOF**

My Universal Forwarder is already configured to monitor /var/log/attack.log into index=attacks

Spl

index=web OR index=attacks src\_ip=\*

| iplocation src\_ip

| top limit=5 Country

#### 8. Correlated sequence with transaction for admin\_user

Use real SSH on Ubuntu

The reasons why we are able to use ssh is because the two vms that kali and ubuntu are set to host only adapter

Where they can speak to each other

On your Kali root terminal, try this against Ubuntu:

Bash

for i in {1..5}; do ssh admin\_user@<192.168.56.103> -o StrictHostKeyChecking=no; done

Note: ip 192.168.56.103 belongs to my ubuntu host only ip

Then one successful login:

ssh admin\_user@<192.168.56.103>

If admin\_user exists on Ubuntu, the. failed ones will show as:

Failed password for admin\_user from <KALI\_IP>

The success will show as:

Accepted password for admin\_user from <KALI\_IP>

My UF on Ubuntu is monitoring /var/log/auth.log into index=auth.

SPL

**index=auth user="admin\_user" (status=failed OR status=success) | transaction user startswith=(status=failed) endswith=(status=success) maxspan=1h | table \_time duration eventcount values(status) values(src\_ip) host**

#### **9. Detect brute-force (>10 failures from same IP within 1 minute)**

**We can use kali to carry out this brute-force attack by using ubuntu host ip**

**ubuntu serve as the victim why splunk serve as the siem tools to monitor the activities.**

**SPL**

**index=auth action="login\_failure" | bin \_time span=1m | stats count as failures by src\_ip \_time | where failures > 10 | sort - failures Add host if present: by src\_ip host \_time.**

#### **10. Dashboard panel — daily unique users for app="sales\_portal" (last 30 days)**

**SPL (panel search) index=app\_logs app="sales\_portal" earliest=-30d@d | timechart span=1d dc(user) as unique\_users Create panel: Save As → Dashboard Panel → New/Existing Dashboard → Visualization: timechart.**

#### **11. Data enrichment with CSV lookup (threat\_list.csv)**

**Upload the CSV Settings → Lookups → Lookup table files → Add new → Upload threat\_list.csv (must include a header like src\_ip). Create a Lookup definition named threat\_list mapped to that file.**

**SPL**

**(return only matches) index=web src\_ip=\* | lookup threat\_list src\_ip OUTPUT src\_ip as matched\_ip | where isnotnull(matched\_ip) | table \_time src\_ip uri\_path status host**

